Unify Learn · Why Each Screen Works This Way

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How to read this

The code and the Front-End Handoff say **what** to build. This doc says **why**: the student problem behind each screen, and what we decided. If you want to change a screen, first check whether the reason still holds.

Each area follows the same shape:

- **The problem:** what students, HOCs or lecturers told us, or what we saw.
- **What the screen does about it.**
- **Decisions:** what we chose and what we rejected.
- **Open:** anything still to settle.

Screen numbers (S01, L01, C01) match the Figma file, page Unify MoodBoard, sections 02 Student app, 03 Contributor and 04 Lecturer app.

Principles behind every screen

1. **3 seconds, one tap.** A student should know what to do next within 3 seconds of opening a screen, and do it in one tap. That's why each screen has one amber button and never two.
2. **We fight early churn.** The biggest risk is students leaving in Weeks 1 to 3, before lecturer materials arrive and contributors turn them into notes. So Home has things to do on day one: a CGPA target, the semester tracker and courses ready to open.
3. **Write like a person.** The voice is friendly, with a little Nigerian warmth, but not heavy Pidgin. This is an academic app, and too much Pidgin dilutes it.
4. **Words that don't hurt.**
    - Never "carryover"; we say PRIORITY. Some students never registered those courses, and the word stigmatises.
    - No shaming on missed streaks.
    - Lecturers aren't all professors, so we never say "Prof".

5. **Notes stay in the app.** There's no "Save as PDF". Students say "save for offline" instead, and the notes stay in Unify, because a PDF removes the reason to come back.
6. **Box Boy is part of the product, not just marketing**, like Duolingo's owl. He appears at success moments, empty states, onboarding, breaks and notifications, so people recognise him across the app and our posts. He's the confident senior who has figured the system out and now guides the students behind him.
7. **Colour means something.** Course blocks are neutral, and colour marks status only. This came from front-end review: "too many aesthetics".
8. **Offline is normal.** Power cuts and data costs are real, so saved courses, notes, quizzes and the attendance pass all work without data.
9. **Measure everything that matters.** Analytics sit at every meaningful step.
10. **The Faculty pays, students don't.** Core access is paid by the Faculty (₦1,000 per student per month), so nothing on a student screen should feel like a paywall at launch. Sponsor slots (the splash footer, streak and certificate moments) are designed so the app still looks complete with no sponsor.

Onboarding (S01–S13)

The problem. Students arrive from a WhatsApp link with no idea what Unify is. If setup feels like a form, they leave. If Home is empty afterwards, they leave too.

What it does. It gets a student from "what is this?" to their own courses on Home in under 2 minutes, and makes the app feel like theirs.

Screen	Why it's there
S01 Splash	Wordmark and "Own your academic journey". The footer says "Powered by Unify" so a sponsor's name can replace it later without a redesign. No LASU logo: we're a partner, not the school's own app.
S02–S05 Slides	Four quick ideas in student language ("It stops now", "We got you"), with Box Boy in each. They sell the reason to sign up before asking for anything.
S06 Sign-up	Google or email. "What should we call you?" is the friendly name on Home, separate from the official ID-card name used for attendance.

Screen	Why it's there
S07 Department	One screen with dropdowns (school, faculty, department, level) instead of four tap-through screens. LASU only for now; other faculties are marked "coming".
S08 Courses	Courses from other levels are allowed and marked PRIORITY, never "carryover".
S09 7b Goal	One of five goals decides where we start them: a quiz, a small weekly task, a worked example, teach-back, or short sessions. It came from the user research segments.
S10 7c Study rhythm	Morning person or night owl, and focus length. These set reminder times. Research showed students' days vary a lot (jobs, businesses, power).
S11 7d Pick your Box Boy	**Now a curved carousel**: the selected avatar is big in the centre, neighbours shrink and fade to the sides, and the student swipes through. Under each one is his persona (e.g. "THE 5.0 SCHOLAR"), a line about him and three traits, plus "Tap him for his story". This makes choosing feel like picking who you are, not just a picture.
S12 Want a little push?	Push permission asked with a reason, not cold. Notifications are how a study app stays alive.
S13 Bag that first class	A win before Home: the first thing to do, ready to tap.

Decisions:

- Students get Box Boy avatars only: no photo uploads and no letter avatars. It keeps the app safe and on-brand, and it removes the "I don't have a good picture" drop-off.
- The matric number and ID-card name are collected later in a short setup, prompted before the first attendance scan, not during onboarding.
- Lecturers and contributors have their own onboarding. There's no "pick your role" screen for students; the current code has one, and it confuses students.

Open: the persona names and traits on the carousel ("The 5.0 Scholar" and the others) are placeholders. Finalise them from the five research segments.

Home, semester week and the CGPA planner (S14–S22)

Home (S14, S19)

The problem. Research showed students lose track of the semester. Week 7 suddenly feels like exam week, and many want a clear path to a 5.0. Home has to give a reason to come back every day, even before notes exist.

What it does:

- **The greeting and department pill** ("300L ELECTRICAL & COMPUTER ENG.") make it theirs.
- **The semester tracker sits under the pill: "Week 2 of 12" with 12 small bars.** It fills week by week. From Week 8 the current bar turns amber and the label reads "Week 9 · exams in 4 weeks". In the last weeks it turns red ("Week 12 · exams next week"). It replaced "Week 1 starts today · first class 8 AM". The class time belongs on the timetable; the hero should show where you are in the semester.
- **CGPA target:** Box Boy holds the target sign. Before a target is set, "Set my target" is the first call to action on Home.
- **Cards, in a fixed order:** Continue (next topic), Today's revision, Next class, My courses. The order follows what the student should do next, not what's newest.
- **"Hide numbers"** lets a student keep their target private on a shared screen.
- **Welcome back (S20)** replaces Home after 5+ days away, so a returning student isn't hit with everything they missed.

Decisions: no wordmark on Home; the Playfair wordmark is for the splash and onboarding only. The amber attendance banner (S58) only appears while a scan is open.

CGPA planner (S15–S18)

The problem. Many students told us they want a clear, honest path to 5.0. The old Grade Planner was dropped in a code re-architecture, and we're bringing it back as Home's first call to action.

Changed on 2 Nov:

- **The student enters their CGPA once** (e.g. 4.00 at 300L), not a grade for every course. A CGPA already combines everything, and typing old grades is tedious and puts people off.
- **Every compulsory course comes pre-ticked as Done.** The student only changes the exceptions:
  - **Priority:** they're retaking it, or it's from an earlier level.
  - **Not yet:** it moved to a later level, or they didn't register it.
- **Units done** are worked out from the ticks, so there's nothing to type. Earlier levels fold into "All done ✓".
- The "Add a course" sheet (S16) asks "Did you take it?" instead of a grade.

How the maths works. Remaining units are the total units minus the units done. For a target T, the GPA needed each remaining semester is (T × total units − current CGPA × units done) ÷ remaining units.

Example: 4.00 with 96 units done and 144 to go. A 4.50 finish needs 4.83 every semester (very difficult). A 4.00 finish needs 4.00 (achievable). The highest possible finish is 4.60.

Honest labels. Each target is tagged out of reach, very difficult, stretch, achievable or comfortable. We'd rather tell the truth than sell a dream.

Privacy. Analytics keep the target band only, never the real CGPA.

Courses, notes, quizzes, revision and streaks (S23–S36)

The problem. Our research found five recurring problems:

- **Scattered materials:** five people asked for one PDF.
- **No study path.**
- **Reading isn't learning.** Students read notes three times and still can't solve a question.
- **Material arrives late,** sometimes in exam week.
- **Lecturers complained** that old questions were too simple.

Course page (S23–S25)

- **Weeks, Outline, Practice.** The semester is broken into the same 12 weeks lecturers teach, so students always know where they are.
- **Locked weeks** open after passing the previous week's quiz (6/10). This keeps students moving in order, without being harsh: the pass mark is reachable.

- **"Notes land by <date>" with Notify me** replaces an empty week. Students see a promise, not a hole.
- **"Pending lecturer review" / "Reviewed by lecturer"** builds trust in student-written notes.
- **Past questions stay locked** until the Faculty approves them, because of copyright.
- **Save for offline** keeps a whole course on the phone, for power cuts and data costs.

Notes (S26, S27)

- **Read → interact → read.** Text cards, formula cards (STIX maths), worked examples with try-then-reveal, quick checks, and a 3-question Pulse check per topic.
- **The card structure is deliberate.** It keeps the eye focused, and formulas, symbols and definitions get extra contrast. Dibble chose this over a flatter style.
- **Report a problem (S27)** goes straight to the contributor who wrote the week, so mistakes get fixed by the person who can fix them.

Quizzes (S28–S32)

- **One question per screen,** with a timer set per quiz by the contributor. Number answers use a number keypad; formula answers use the maths keypad (S31) and are checked as expressions, so $0.296\rho V^3$ and $8/27 \cdot \rho \cdot V^3$ both count.
- **Theory questions** show the model solution after the final answer, so students compare their working, not just the answer.
- **Copying and the copy menu are blocked** so questions can't be pasted into AI. A web app can't fully block screenshots, so the copy says "discouraged", not "impossible".
- **Results (S32)** link wrong answers back to the topic ("Worth another look").
- **Week complete (S33)** is a Box Boy celebration that leads straight into the next week.

Revision and streaks (S34–S36)

- **Spaced revision** across all courses brings questions back after 1, 3, 7, 14 and 30 days. It's capped at 10 a day, so it never feels like a second job.
- **Streak:** a day counts if the day's revision or a class topic is done. A day runs 4 AM to 4 AM (night readers). Weekends never break it, and a missed day resets quietly, with no shaming.

Open: Focus mode (Pomodoro reading blocks, using the length from onboarding) is a later pro feature.

Timetable, Class tab, Me and Settings (S37–S53)

Timetable (S37–S42)

The problem. Students get class information from the HOC on WhatsApp. Changes get buried under hundreds of messages, and students go to an 8 AM class that isn't happening.

What it does:

- **The week view is the default**, because students plan by the week. Day view is one tap away.
- **Neutral blocks.** Colour only marks status: red for cancelled, dashed for moved or not confirmed, a grey chip for a course from another level. The first version gave every course a pastel colour, and the front-end review called it "too many aesthetics". In a tool everyone uses daily, familiar and calm beats colourful.
- **"Changes this week"** sits under the grid, signed by the HOC, so students see who changed what.
- **Who sets it:** the Faculty's Monday–Friday timetable is the base for everyone. The HOC adjusts it for their class, and a lecturer's change wins. Lecturers never set a student timetable from scratch.
- **Lecturer check-in (S39):** "Is the lecturer in class?" A class counts as confirmed when the lecturer taps Confirm or 3 classmates confirm, which saves wasted trips.

Class tab (S42–S43)

- **Announcement-only, no chat,** like a locked WhatsApp group. Chat creates noise and moderation work. The point is that important notices never get buried.
- **Official posts are marked "From Dr. ___"** so students know what's real.
- **Late registrants see the full history,** so nobody misses the first-week announcement.
- The tab badge is a round red count, easy to spot.

Me and Settings (S45–S53)

- **Me (S45)** is the student's corner: Box Boy, streak, saved courses and notifications. The **class rep version (S46)** adds the "Class rep tools" card, so HOCs use the same app and don't need a separate login.
- **Settings (S48–S49)** is separate from Me so Me stays simple: notification times, quiet hours, attendance setup.

- **Become class rep (S50–S51):** the request goes to the Unify team and the student sees "waiting", because anyone claiming to be HOC could post to a whole class.
- **Offline (S52):** says what still works, rather than showing an error.
- **Lecturer profile (S53):** what a student sees about a lecturer: photo or Box Boy, bio, how to reach them. Lecturers choose what to show.

Class rep (HOC) tools and contributors (S54–S56, C01–C06)

Head of Class (S54–S56)

The problem. HOCs already do this work for free, on WhatsApp and paper: timetable changes, materials, chasing lecturers. Unify was also leaning on them without giving anything back.

What it does:

- **Post to class (S54):** the same post types as lecturers, marked as from the HOC.
- **Edit timetable (S55):** shows clashes and classes that aren't confirmed yet.
- **Contributors (S56):** who writes each course, whether they're behind, and "Upload notes from your lecturer". HOCs are our bridge to lecturers' materials.

Decisions:

- **There can be 1 to 3 HOCs per class.** The tools make their existing work faster, not add more.
- **The Unify team approves HOCs,** because a HOC can post to hundreds of students.
- **HOCs confirm which lecturer teaches a course** (S62), so a stranger can't claim a class.
- **HOCs are recognised, not just used.** They're one of three groups in the reward programme, alongside contributors and ambassadors.

Contributors (C01–C06)

The problem. Lecturer materials are slides and PDFs, not study-ready notes. Students who aced a course can explain it better than the slides, and they deserve to be paid for it.

What it does. The dashboard (C01) shows what's due this week, days left on each 7-day deadline, and student reports to fix. Studio (C02–C06) goes Compose → Review → Live:

- AI Generate, Manual blocks, or External AI
- the quiz builder with a timer and pass mark

- review, then live

Decisions:

- **The AI engine is optional, not a dependency.** The note format is deterministic, so a week can always be built by hand.
- **At most 2 courses per level per contributor,** so quality stays high and the work is spread out.
- **A week goes live after the Unify team checks it within 24 hours.** It then shows "Pending lecturer review" until the lecturer marks it reviewed.
- **More than 3 days late lets admin reassign the week,** so students aren't left waiting.
- **Contributors are paid per approved week.** Unify should stop relying on free student help.

Long term: every level writes for the level behind it. 400L writes 300L notes, 300L writes 200L, and so on.

Attendance in person (S57–S63)

The problem. Classes of up to 750 spend 35+ minutes on name-calling or paper lists, and proxy signing is common. Attendance isn't in the Faculty proposal; it's an add-on, built after Week 8.

What it does:

- **Only the lecturer can open a scan,** from Home at class time. HOCs of every department in that class get a push.
- **Each student has one pass (S59) for all their courses:** a QR code that changes every 30 seconds.
- **HOCs and/or the lecturer scan passes at the door (S61)** as students leave, which takes about 5 minutes. The scanner shows the student's Box Boy, name, level and department with a tick, or a clear reason it failed.
- **Each student sees their own percentage (S63).** Below 75% they get a gentle nudge to check their department's rule.

Decisions:

- **The pass is signed by Unify's server and tied to the student's phone,** so it can't be forged or used on another phone. A screenshot dies within about 2 minutes, and everything works offline.

- **The HOC can type in a matric number** for a dead phone, marked "typed in" so the lecturer can see it.
- **Known limit:** nothing offline proves someone is physically in the room. The people side covers it: a HOC at the door, an unannounced scan time, a 10-name spot check, and "scanned by" on every row.
- **Wording is "Came to 1 of 4 classes held",** never a bare percentage without context.

Open: verified class lists (discussed 31 Oct, parked)

The problem. Students type their own names, so "Kendi Blossom" becomes "Blossom Kendi" and confuses the lecturer's register and exports. Registers need one official source.

What we agreed:

1. **The HOC uploads the class list,** a photo of the department's official list. Unify reads the names and matric numbers, and the HOC checks a preview before publishing. This becomes the source of truth for the lecturer, the HOC and students.
2. **The matric number is the key, not the name.** Registers, PDFs and exports always print the official name from the list, including the middle name. The student's display name in the app can be anything.
3. **Students claim their entry by signing in with their LASU school email** ("Continue with Google", only st.lasu.edu.ng accounts). The email format is firstname.lastname + matric number @st.lasu.edu.ng, so Unify reads the matric number from it and matches the list. That's one tap, once, forever, with no code to type.
4. **Typing the email into a box is not enough.** The format is guessable, so anyone could claim a classmate. Signing in proves they own the inbox.
5. **Unverified students can still learn** (courses, notes, quizzes, streaks). Only official records need verification: attendance, registers and rewards or payments.
6. **Fallback:** students who can't use their school email request a claim, and the HOC clears these in a weekly batch, not one by one for 350 students.
7. **Students who never show up** stay unclaimed on the list and appear as absent. Nobody needs to chase them.

Still open:

- **A forgotten school-email password.** Likely answer: LASU ICT resets it, and they learn unverified until then.

- **Students who signed up with a normal Gmail** link their school email later from Settings. The flow isn't designed yet.
- **Three checks before building:**
  1. Does the email pattern hold for every department?
  2. Does LASU's Google allow "Sign in with Google" for outside apps?
  3. How many students have activated their school email?
- **Data protection:** class lists are personal data. Students only ever see their own entry; full lists are for the HOC, the course lecturer and admin.

Online classes (S64–S70, L18–L20)

The problem. Lecturers already teach some classes on Google Meet, Zoom, Telegram or WhatsApp. Links get lost in group chats, attendance is guesswork, students drop out on bad networks with no way to say so, and long classes run with no break.

What it does. Unify doesn't host video. It wraps whatever link the lecturer already uses, so students join through Unify and attendance counts automatically.

- **Join through Unify (S64–S65).** The class page says it's online and which room is yours. Join opens 5 minutes before class. Joining is check-in 1.
- **2 of 3 check-ins = present (S66–S67).** Two random "Still here?" prompts, each one tap within 2 minutes. Two of three, not three of three, because networks drop and we don't want to punish bad data.
- **Lost connection (S69).** One tap and a reason (network, light, data, phone died). It shows on the register next to their name, and the lecturer decides whether to count it. It's honest on both sides.
- **Breaks (S68).** The lecturer can start a 10-minute break and check-ins pause. Students asked for a break in long online classes.
- **Class done (S70)** shows the attendance result, then points to this week's notes and the recording.
- **The lecturer (L18–L20)** moves a class online in one screen, runs it live with counts and "Send a check-in now", then gets a register split into present, partly there, lost connection and didn't join.

Decisions:

- **Rooms for big classes.** Free Meet and Zoom rooms hold 100 people, so a 400-student class gets 4 rooms. The class is split by surname, matric number or evenly. Each student only sees their own room, and a full room moves them to the next.
- **We're honest about what Unify knows.** It records the Join tap and the check-in answers, not presence inside Meet. An optional upload of Meet's participant list is the cross-check.
- **The online register feeds the same attendance count** as in-person classes.

Open: the lecturer can only be in one room. Do rooms run at the same time (the lecturer teaches in Room 1 and HOCs share the screen elsewhere, or one Telegram or YouTube stream with no viewer limit), or as batches at different times? The design supports an optional start time per room either way.

Lecturer app (L01–L17)

The problem. Lecturers repeat the same work for every class: sending materials, announcing changes, answering "is there class today?". One lecturer can teach a shared course to 742 students across 6 departments, plus a single class elsewhere, at any level from 100L to 500L.

What it does. It's its own product that makes the lecturer's job lighter: **send it once**, and every class taking the course gets it.

Screen	Why it's there
L01–L06 Onboarding	"SEND IT ONCE" sets the promise. Lecturers search any department and level, because they teach across both. Photo, bio and contact details are optional: lecturers choose what students see.
L07–L08 Home	One main button at a time. Before class it's **Confirm class**, with a "Running late or cancelling?" link. At class time it's **Open attendance scan**. Students see "Lecturer confirmed", which ends the "is there class?" messages.
L09 Timetable	Class timetables set by HOCs, plus a private "My own" (office hours, duties). A lecturer's change to a class wins.

Screen	Why it's there
L10–L12 Course page	Weeks, Report and Attendance, with a department switch for shared courses. Up to 3 co-teaching lecturers.
L13 Review a week	The note exactly as students see it, with "Mark as reviewed" or "Needs changes". Lecturer review is what makes student-written notes trustworthy.
L14 Send materials	Upload once for a course and week. It goes to Unify, not straight to students, and contributors turn it into notes within 7 days.
L15 Post to students	Pick the course, then all or some departments, then the message. Announcement-only, and it confirms before sending to hundreds.
L16 Attendance list	"Came to 1 of 4 classes held". Rows show level, department and matric number, because names repeat. Saves as PDF or Excel.
L17 Me	Profile, courses "Waiting for Head of Class", notifications.

Decisions:

- **Reports show counts only, never named students.** Lecturers see engagement, not surveillance.
- **The HOC confirms a lecturer's claim to a course,** so the wrong person can't post to a class.
- **Lecturer screens use bigger text (17px), high contrast and words on every button.** Many lecturers aren't app-native.
- **Never "Prof":** not every lecturer is a professor.

Caution: we haven't interviewed any lecturers yet, so this whole side is built on assumptions. Build what the Faculty proposal promises first (Send materials, Post, Review, Report), keep the rest easy to change, and interview lecturers before Week 8.

Payment: a per-course review honorarium is being considered (₦10,000 per course per review proposed, still under discussion).

States, offline and notifications (S71–S76)

States (S71–S76)

Why. An error that only says "Error 500" makes students think the app is broken, and they leave. Every state says what happened and what to do next. Box Boy softens the moment. The states covered:

- something went wrong, with "open saved notes"
- wrong verification code, with tries left and a resend timer
- course search with no results, with "check the code on your timetable"
- pass needs a refresh
- quiz saved offline
- a lecturer with no confirmed courses yet

Offline

Why. Power cuts and data costs are part of student life here. Offline is treated as normal, not as an error:

- saved courses, notes, the attendance pass and quiz answers all work without data
- anything done offline syncs later
- things that work offline show a "Saved offline" chip, never a red banner

Push notifications (the board under States)

Why. It's a study app, and without strong notifications it gets forgotten. But too many notifications get the app muted. The rules:

- **The morning brief at 7 AM** (or the student's chosen time) has one call to action. It's personal: name, progress, today's classes.
- **At most 3 a day.** Class changes (cancelled, moved, test, moved online) and online check-ins always go through.
- **Quiet from 10 PM to 6 AM** unless the student chose "Night owl".
- **Every notification opens the exact screen it's about.**
- **The icon is Box Boy**, not the logo. It's friendlier, and it's recognisable.

Open questions and parked decisions

To decide

Question	Why it matters	Leaning
Verified class lists: school-email sign-in (see Attendance)	Registers and exports need one official name per matric number	Agreed in principle; the forgotten-password and linking flows are still open
Online classes: rooms at the same time or batches?	The lecturer can only be in one room	Design supports both; LASU practice decides
Box Boy persona names and traits on the carousel	Students choose who they identify with	Draft from the five research segments
Attendance pass lifetime: 1 day or up to 7 days	Students without data for days still need a working pass	Not decided
Devices per student per semester	Phone theft, shared phones vs proxy risk	Not decided
Contributor pay rate and lecturer review honorarium	Fairness and budget	₦10,000 per course per review proposed
When past questions can unlock	Faculty copyright approval	Waiting on the Faculty
Exact department names (Industrial & Production?)	Shown everywhere in onboarding and filters	Confirm all 6

Parked on purpose

- **Dark mode:** option A is chosen. Built with colour roles so it's a switch later.
- **Chat** between lecturers, HODs and the Dean.
- **Communication, Opportunity and Accessibility gaps:** after the learning product is proven.
- **Focus mode and other pro features:** after Week 8.
- **HOD and Dean panels:** due by Week 8, design still being refined, so not in this code drop.

- **Desktop layouts:** not in this code drop. The phone layout comes first.
- **Face-off (1v1 game):** a mock design only, not for build yet.

What changed on 2 Nov

- The Home hero shows the semester week tracker instead of "Week 1 starts today · first class 8 AM".
- The CGPA planner asks for the CGPA once and course status (Done, Priority, Not yet), not grades.
- "Pick your Box Boy" is a curved, fading carousel with a persona and traits under each avatar.